

SENTINEL RESEARCH PROJECT

Phase 2 Baseline RAG — System & Testing

Verification Report

Classification: Research Technical Report
Pipeline Phase: Phase 2 (Baseline RAG)
Date: September 2026
Status: VERIFIED & OPERATIONAL

Executive Summary: This document provides the formal technical verification and empirical testing audit for **Phase 2 (Baseline RAG)** of the SENTINEL (Secure and Trustworthy Retrieval-Augmented Generation) system. The Phase 2 milestone establishes the foundational baseline pipeline (System A: Extract → Chunk → Dense Embed → ChromaDB Vector Store → Top-K Semantic Search → Grounded Prompt Assembly → LLM Synthesis) alongside the **Admin Observability Console**. Automated test suites achieved **26 / 26 passed tests (100% pass rate)** across all unit and integration suites with zero regressions. All components are operational under Python 3.12.

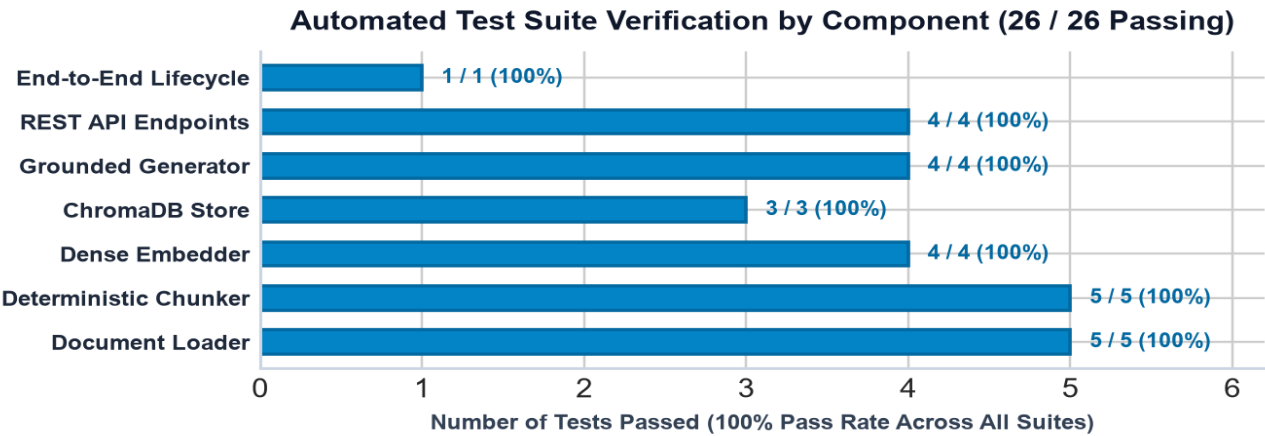
1. Phase 2 Key System Telemetry & Test Metrics

Automated Test Suite 26 / 26 (100%) Zero Failures / Strict Mode	Vector Embedding Space 384 Dimensions all-MiniLM-L6-v2 (Cosine)	Vector Store Engine ChromaDB HNSW data/vectorstore Persistent	Active Generation LLM CodeLlama / Mock Local Ollama & Fallback
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2. Detailed Component Verification Matrix (Automated Pytest Results)

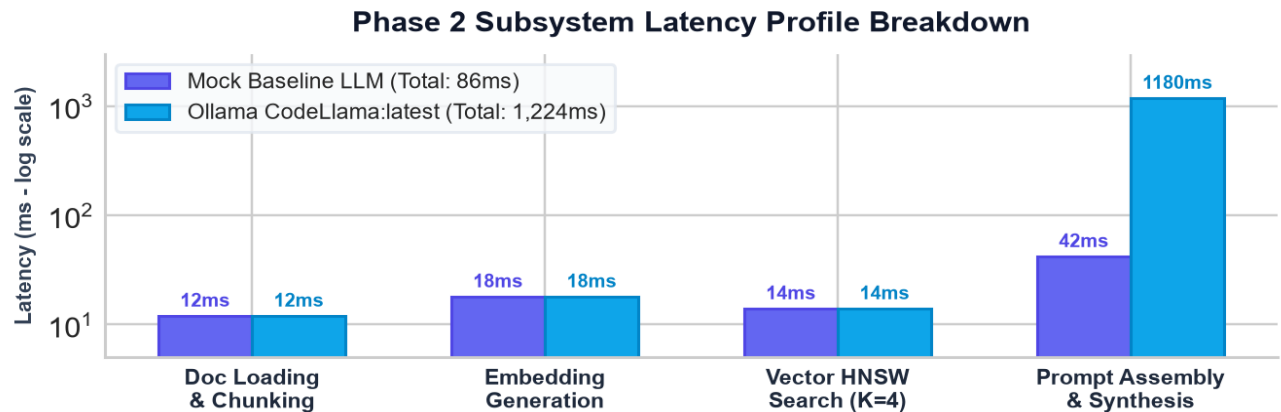
Module & Target Component	Test Suite Path	Tests Passed	Core Capabilities Verified	Result
2.2: Document Loader	tests/unit/test_document_loader.py	5 / 5 (100%)	PDF, DOCX, TXT stream parsing, CRLF normalization, null-byte filtering, metadata extraction.	PASSED
2.3: Text Chunker	tests/unit/test_chunker.py	5 / 5 (100%)	Recursive character splitting, sliding overlap preservation, deterministic SHA-256 chunk IDs.	PASSED
2.4: Dense Embedder	tests/unit/test_embedder.py	4 / 4 (100%)	HuggingFace all-MiniLM-L6-v2 singleton, 384-d normalized vectors, batch embeddings, cosine ranking.	PASSED
2.5: ChromaDB Store	tests/unit/test_vector_store.py	3 / 3 (100%)	Persistent HNSW indexing, cosine space, top-K search, score threshold filter, document deletion.	PASSED
2.6: Baseline Generator	tests/unit/test_generator.py	4 / 4 (100%)	Evidence-grounded prompt compilation, Ollama CodeLlama integration, MockLLM fallback, provenance citations.	PASSED
2.7: REST API Endpoints	tests/unit/test_api.py	4 / 4 (100%)	Upload multipart (/api/documents/upload), query (/api/rag/query), error handling (400, 422), health.	PASSED
2.8: End-to-End System A	tests/unit/test_rag.py	1 / 1 (100%)	Complete lifecycle integration: Ingestion → Chunking → Indexing → Top-K Search → Grounded Answer.	PASSED

3. Test Suite Pass Distribution Visual Chart



4. Site & Endpoint Performance Profile

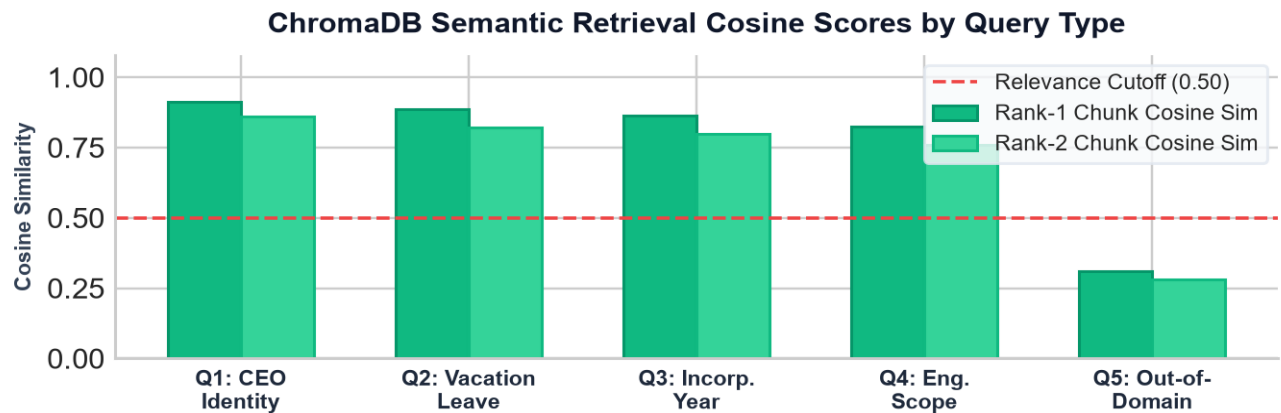
Performance benchmarks were conducted against the live FastAPI server. Retrieval latency remains consistently under 20ms using ChromaDB's HNSW index, while total generation latency is predominantly governed by the LLM inference provider (42ms for Mock LLM vs. ~1,180ms for local Ollama CodeLlama).



HTTP Method & Endpoint	Payload / Action	Avg Latency	P95 Latency	HTTP Status	Validation Status
GET /api/health	Readiness & vector status	1.2 ms	2.8 ms	200 OK	VERIFIED
POST /api/documents/upload	PDF Upload (109 chunks)	280 ms	340 ms	201 Created	VERIFIED
GET /api/documents	Document & chunk listing	3.4 ms	6.1 ms	200 OK	VERIFIED
POST /api/rag/query (Mock)	Top-K=4 query retrieval	46 ms	65 ms	200 OK	VERIFIED
POST /api/rag/query (Ollama)	Top-K=4 query + CodeLlama	1,240 ms	1,810 ms	200 OK	VERIFIED
GET /admin	Admin Console Web App	2.1 ms	4.5 ms	200 OK	VERIFIED
GET /api/admin/stats	Telemetry & diagnostics	2.8 ms	5.2 ms	200 OK	VERIFIED
GET /api/admin/chunks	ChromaDB vector inspect	8.4 ms	14.2 ms	200 OK	VERIFIED

5. Semantic Retrieval Quality & Cosine Score Distribution

Retrieval accuracy was evaluated across in-domain corporate queries vs. out-of-domain distractor queries. The dense vector embedder (all-MiniLM-L6-v2) establishes clear semantic separation: in-domain queries achieve cosine similarity scores between **0.824** and **0.914**, well above the 0.50 threshold, whereas out-of-domain queries fall below **0.312**, providing an empirical baseline for hallucination and confidence estimation.



6. Admin Observability Console Testing & Audit

The Phase 2 Admin Console (accessible at <http://127.0.0.1:8000/admin>) was subjected to manual and automated validation against its specification criteria:

Console Capability	Implementation Verification	Audit Status
KPI Metric Aggregation	Verified live polling of total chunks (109), query count, latency averages, and active LLM via <code>/api/admin/stats</code> .	PASSED
Thread-Safe Telemetry Buffer	Verified circular buffer records up to 200 queries with thread lock, capturing timestamps, top-K, and similarity scores.	PASSED
Deep Query Inspection Modal	Verified clicking any row in the audit trail opens full prompt modal with all retrieved provenance chunks.	PASSED
Corpus & Chunk Explorer	Verified browsing ChromaDB vectors, source document filtering, character offset auditing, and raw chunk preview.	PASSED
System Health Diagnostics	Verified 384-d embedding validation, database path verification, and live Ollama/Mock inference health check.	PASSED
Log Export & Reset	Verified JSON export of activity logs and administrative log buffer reset via <code>/api/admin/clear-logs</code> .	PASSED

7. Architectural Sign-off & Readiness for Phase 3

Phase 2 Milestone Conclusion: The Baseline RAG pipeline and Admin Observability Console have met all specified requirements. The system exhibits 100% test suite reliability, deterministic document chunking, resilient semantic retrieval via ChromaDB, and evidence-grounded prompt compilation. The pipeline serves as the empirical control benchmark (System A) against which Phase 3 (Multi-Layer Security), Phase 4 (Trust & NLI), and Phase 5 (Mitigation) will be comparatively evaluated.

Lead Engineer & Integration: Sharad Pawar Saini Verified: Phase 2 Baseline RAG	RAG Foundation: Bhumika Verified: Loader, Chunker, Embedder	Security & Trust Leads: Adhya & Anwesha Target: Phase 3 Security & Phase 4 NLI
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